Q1 JEE Main 2020 - 2 September (Morning)

For the following Assertion and Reason, the correct option is
Assertion (A): When Cu (II) and sulphide ions are mixed, they react together extremely quickly to give a solid.
Reason (R): The equilibrium constant of
$\mathrm{Cu}^{2+}(\mathrm{aq})+\mathrm{S}^{2-}(\mathrm{aq}) \rightleftharpoons \mathrm{CuS}(\mathrm{s}) \mathrm{is}$
high because the solubility product is low.

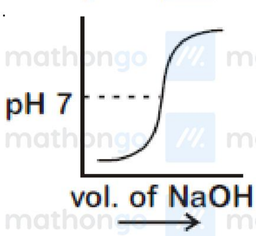
(A) (A) is false and (R) is true.
(B) Both (A) and (R) are true but (R) is not the explanation for (A).
(C) Both (A) and (R) are true and (R) is the explanation for (A).
(D) Both (A) and (R) are false.

Q2 JEE Main 2020 - 3 September (Morning)

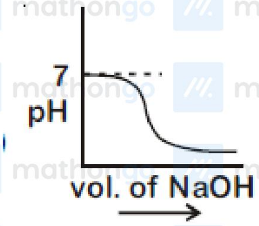
An acidic buffer is obtained on mixing
(A) 100 mL of 0.1 M HCl and 200 mL of 0.1 M NaCl
(B) 100 mL of 0.1 M HCl and 200 mL of 0.1 M $\mathrm{CH_3COONa}$
(C) $100 \mathrm{mL}$ of $0.1 \mathrm{M} \mathrm{CH_3COOH}$ and $100 \mathrm{mL}$ of $0.1 \mathrm{M} \mathrm{NaOH}$
(D) $100 \mathrm{mL}$ of $0.1 \mathrm{M} \mathrm{CH_3COOH}$ and $200 \mathrm{mL}$ of $0.1 \mathrm{M} \mathrm{NaOH}$

Q3 JEE Main 2020 - 3 September (Evening)

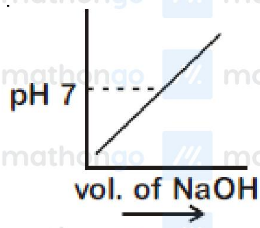
$100 \mathrm{mL}$ of $0.1 \mathrm{M} \mathrm{HCl}$ is taken in a beaker and to
it $100 \mathrm{mL}$ of $0.1 \mathrm{M} \mathrm{NaOH}$ is added in steps of $2 \mathrm{mL}$ and the pH is continuously measured. Which of the following graphs correctly depicts the change in pH?



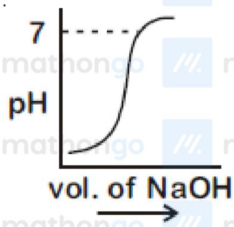
(A)



(B)



(C)



(D)

Q4 JEE Main 2020 - 6 September (Morning)

Arrange the following solutions in the decreasing order of pOH
(A) $0.01 \mathrm{M HCl}$
(B) $0.01 \mathrm{M NaOH}$
(C) $0.01 \mathrm{M CH_3COONa}$
(D) $0.01 \mathrm{M NaCl}$

(A) $(B)>(C)>(D)>(A)$
(B) $(\mathrm{A})>(\mathrm{D})>(\mathrm{C})>(\mathrm{B})$
(C) $(\mathrm{A})>(\mathrm{C})>(\mathrm{D})>(\mathrm{B})$
(D) $(B)>(D)>(C)>(A)$

Q5 JEE Main 2020 - 6 September (Evening)

If the solubility product of $\mathrm{AB_2}$ is $3.20 \times 10^{-11} \mathrm{M}^3$, then the solubility of $\mathrm{AB_2}$ in pure water is $\times 10^{-4} \mathrm{mol L^{-1}}$. [Assuming that neither kind of ion reacts with water]

Q6 JEE Main 2020 - 7 January (Morning)

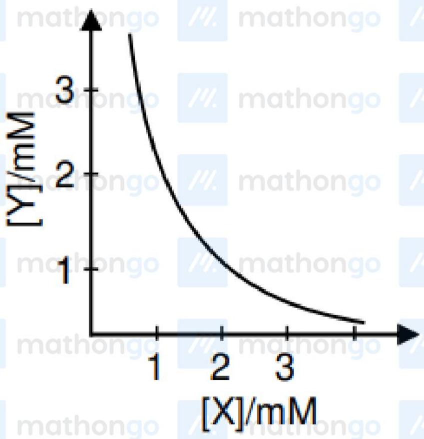
Two solutions, A and B, each of $100\ L$ was made by dissolving $4\ g$ of $NaOH$ and $9.8\ g$ of H_2SO_4 in water, respectively. The pH of the resultant solution obtained from mixing $40\ L$ of solution A and $10\ L$ of solution B is

Q7 JEE Main 2020 - 7 January (Evening)

3 gram of acetic acid is mixed in $250\ mL$ of $0.1\ M\ HCl$. This mixture is now diluted to $500\ mL \cdot 20\ mL$ of this solution is now taken in another container. $\frac{1}{2} mL$ of $5\ M\ NaOH$ is added to this. Find the pH of this solution. $(\log 3 = 0.4771,\ pK_a = 4.74)$
Multiply your answer with 100

Q8 JEE Main 2020 - 8 January (Morning)

The stoichiometry and solublity product of a salt with the solubility curve given below is, respectively



(A) $XY,\ K_{sp} = 2 \times 10^{-6}$
(B) $XY_2,\ K_{sp} = 4 \times 10^{-9}$
(C) $X_2Y,\ K_{sp} = 9 \times 10^{-9}$
(D) $XY_2,\ K_{sp} = 1 \times 10^{-9}$

Q9 JEE Main 2020 - 8 January (Evening)

For the following Assertion and Reason, the correct option is :
Assertion : The pH of water increases with increase in temperature.

Reason : The dissociation of water into H^{+} and OH^{-} is an exothermic reaction.

(A) Both assertion and reason are correct and reason is correct explanation of assertion.
(B) Both assertion and reason are correct and reason is not correct explanation of assertion.
(C) Assertion is true & reason is false.
(D) Both assertion and reason are incorrect.

Q10 JEE Main 2020 - 9 January (Morning)

The K_{sp} for the following dissociation is 1.6×10^{-5}

$$PbCl_2(s) \rightleftharpoons Pb^{2+}_{(aq)} + 2Cl^{-}_{(aq)}$$

Which of the following choices is correct for a mixture of $300\ mL\ 0.134\ M\ Pb(NO_3)_2$ and $100\ mL\ 0.4\ M\ NaCl$?

(A) $Q > Ksp$
(B) $Q < Ksp$
(C) $Q = Ksp$
(D) Not enough data provided

Q11 JEE Main 2020 - 9 January (Evening)

Amongst the following, the form of water with the lowest ionic conductance at $298\ K$ is:

(A) distilled water
(B) sea water
(C) water from a well
(D) saline water used for intravenous injection

Answer Key

Q1 (C)	**Q2** (B)	**Q3** (A)	**Q4** (B)
Q5 (2)	**Q6** (10.6)	**Q7** (522)	**Q8** (B)
Q9 (D)	**Q10** (A)	**Q11** (A)	

Q1

$\mathrm{K_{sp}}$ value of CuS is very low $\simeq 10^{-36}\left(3.6\times 10^{-36}\right)$
due to low $\mathrm{K_{sp}}$ value $\mathrm{Cu^{+2}}$ ion gets precipitated very quickly even with very low concentration of S^{-2} ion

$\mathrm{CuS(s)} \rightleftharpoons \mathrm{Cu^{+2}} + \mathrm{S^{-2}}$

$\mathrm{K_{sp}} = \left[\mathrm{Cu^{+2}}\right]\left[\mathrm{S^{-2}}\right]$

$\mathrm{Cu^{+2}} + \mathrm{S^{-2}} \xrightarrow{\mathrm{K_{eq}}} \mathrm{CuS(s)}$

$\mathrm{K_{eq}} = \frac{1}{\mathrm{K_{sp}}} = \frac{1}{3.6\times 10^{-36}}$

$= \frac{10^{36}}{3.6}$

Due to high value of K (equilibrium constant) CuS precipitated quickly.

Q2

	HCl	+ CH_3COONa	$\rightarrow$ CH_3COOH	+ NaCl
Initial	100×0.1 = 10 mmol	200×0.1 = 20 mmol		
Final moles	0	10 mmol	10 mmol	10 mmol

$\mathrm{CH_3COOH}$ and $\mathrm{CH_3COONa}$ both are present.
Both form acidic buffer.

Q3

For titration between HCl and NaOH, pH at equivalence point is found to be 7 .

Q4

(A) $0.01\mathrm{M HCl}$

$\left[\mathrm{H^+}\right] = 10^{-2}, \mathrm{pH} = -\log 10^{-2} = 2$

$\mathrm{pOH} = 14 - 2 = 12$

(B) $0.01\mathrm{M NaOH}$

$\left[\mathrm{OH^-}\right] = 10^{-2}, \mathrm{pOH} = -\log[\mathrm{OH}]$

$= 2$

(C) $0.01 \mathrm{MCH_3COONa}$
$\mathrm{pH} = 7 + \frac{1}{2}[\mathrm{pK_a} + \log 0.01]$
$\mathrm{pH} > 7 \Rightarrow \mathrm{pOH} < 7$
(D) $0.01 \mathrm{MNaCl}, \mathrm{pH} = 7, \mathrm{pOH} = 7$
Order of pOH value $A > D > C > B$

Q5

$$\mathrm{AB_2} \rightleftharpoons \underset{\mathrm{s}}{\mathrm{A}^{2+}}(\mathrm{aq}) + \underset{2\mathrm{s}}{2\mathrm{B}^{-}}(\mathrm{aq})$$

$$\mathrm{K_{sp}} = 4\mathrm{s}^3 = 3.2 \times 10^{-11}$$

$$\Rightarrow \mathrm{s}^3 = 8 \times 10^{-12}$$

$$\mathrm{s} = 2 \times 10^{-4}$$

Q6

$$M_{H_2SO_4} \Rightarrow \frac{9.8}{98 \times 100} = 10^{-3}$$

$$M_{NaOH} \Rightarrow \frac{4}{40 \times 100} = 10^{-3}$$

$$= \frac{40 \times 10^{-3} - 10 \times 10^{-3} \times 2}{50} = \frac{20}{50} \times 10^{-3}$$

$$[OH^{-}] = \frac{2}{5} \times 10^{-3}$$

$$pOH = \frac{2}{5} \times 10^{-3}$$

$$pOH = 3.397$$

$$pH = 10.603$$

Q7

m mole of acidic acid in 20 $mL = 2$
m mole of HCl in 20 $mL = 1$
m mole of $NaOH = 2.5$

$$\begin{array}{ccccccccc} CH_3COOH & + & NaOH\,(\text{remaining}) & \rightarrow & CH_3COONa & + & \text{water} \\ 2 & & 3/2 & & 0 & & 0 \\ 0.5 & & 0 & & 3/2 & & - \end{array}$$

$$pH = PK_a + \log\frac{3/2}{2}$$
$$= 4.74 + \log 3$$
$$= 4.74 + 0.48 = 5.22$$

Final ans : $5.22 \times 100 = 522$

Q8

$$XY_2(s) \rightarrow X^{+}(aq) + 2Y^{-}(aq)$$
$$K_{sp} = [X^{+}][Y^{-}]^2$$
$$or\ K_{sp} = 10^{-3} \times (2 \times 10^{-3})^2$$

Q9

Theory Based

Q10

$$Q = \left[Pb^{2+}\right]\left[Cl^{-}\right]^2$$
$$= \frac{300 \times 0.134}{400} \times \left[\frac{100 \times 0.4}{400}\right]^2$$
$$= \frac{3 \times 0.134}{4} \times (0.1)^2$$
$$= 0.105 \times 10^{-2}$$
$$= 1.005 \times 10^{-3}$$
$$Q > Ksp$$

Q11

Theory based